

**Industrial Internship Report on
" Agriculture Crop Production and Yield Prediction using Machine
Learning"
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Executive Summary

This report presents the work completed during the Industrial Internship conducted by Upskill Campus in collaboration with UniConverge Technologies Pvt. Ltd. (UCT). The internship focused on solving a real-world problem using Machine Learning and Data Analytics techniques.

The project, titled **Agriculture Crop Production and Yield Prediction using Machine Learning**, aims to predict agricultural crop yield and future crop production in India using historical agricultural datasets. Agriculture plays a vital role in India's economy, and accurate prediction of crop production can help farmers, researchers, and policymakers make informed decisions.

The project involved collecting agricultural datasets, preprocessing the data, performing exploratory data analysis, engineering new features, and training multiple machine learning models. Several regression algorithms were evaluated, and the best-performing models were selected based on performance metrics such as R^2 Score, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE).

A user-friendly Streamlit web application was also developed to allow users to perform crop yield and production predictions interactively.

This internship significantly enhanced my understanding of Machine Learning, Data Analysis, Feature Engineering, Model Evaluation, and Full Stack Deployment using Streamlit. It also provided valuable exposure to solving real-world industrial problems using Artificial Intelligence.

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1. Preface

Industrial internships bridge the gap between academic learning and practical industrial applications. This internship offered an opportunity to understand how Machine Learning can be applied to solve real-world agricultural challenges.

Over six weeks, I worked on designing and implementing a complete machine learning solution capable of predicting agricultural crop yield and production using historical agricultural datasets. The project involved every stage of the machine learning lifecycle, including data collection, preprocessing, exploratory data analysis, feature engineering, model training, evaluation, and deployment.

The internship was carefully planned with weekly deliverables that encouraged continuous learning and practical implementation. Throughout the internship, I learned how to manage datasets, compare multiple regression algorithms, evaluate model performance, and deploy predictive models using Streamlit.

I sincerely thank **Upskill Campus**, **The IoT Academy**, and **UniConverge Technologies Pvt. Ltd.** for providing this excellent learning opportunity. Their structured guidance, resources, and industrial exposure greatly contributed to my professional development.

I would also like to thank my mentors and faculty members for their continuous support throughout the internship.

This internship strengthened my confidence in applying Artificial Intelligence and Machine Learning techniques to solve real-world problems and has motivated me to pursue further research in Data Science and Predictive Analytics.

2. Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



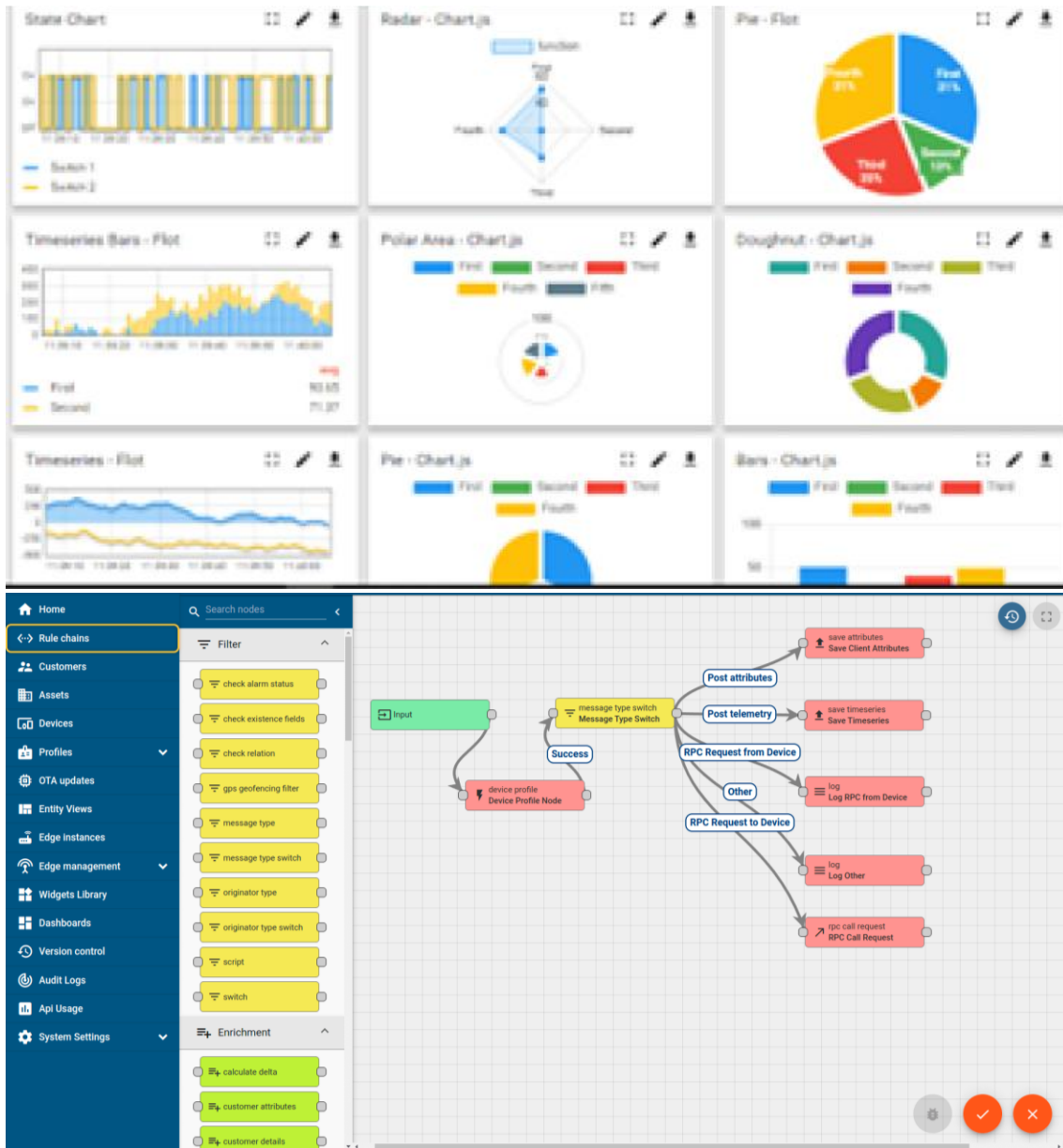
i. UCT IoT Platform

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



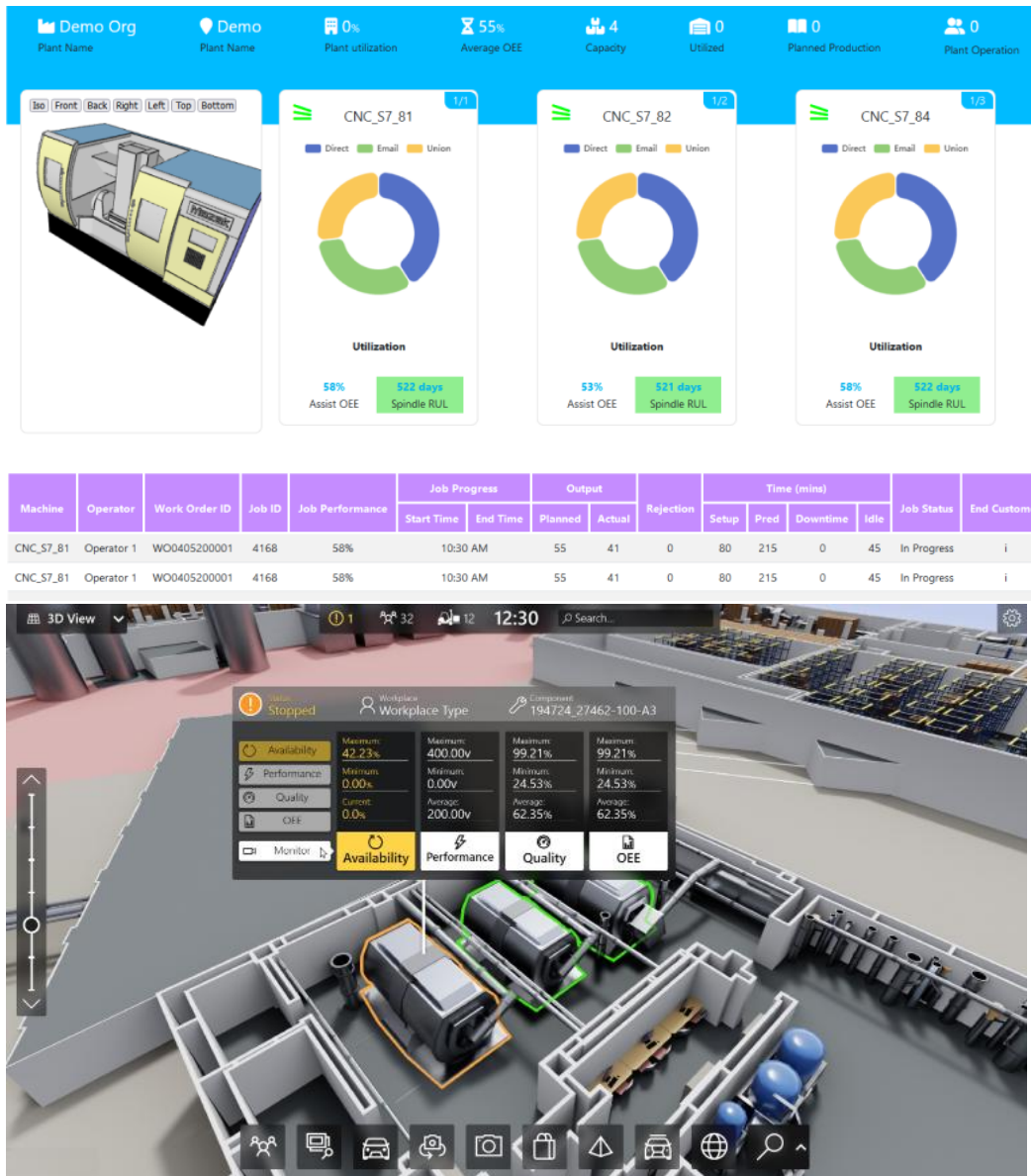
ii. Smart Factory Platform

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

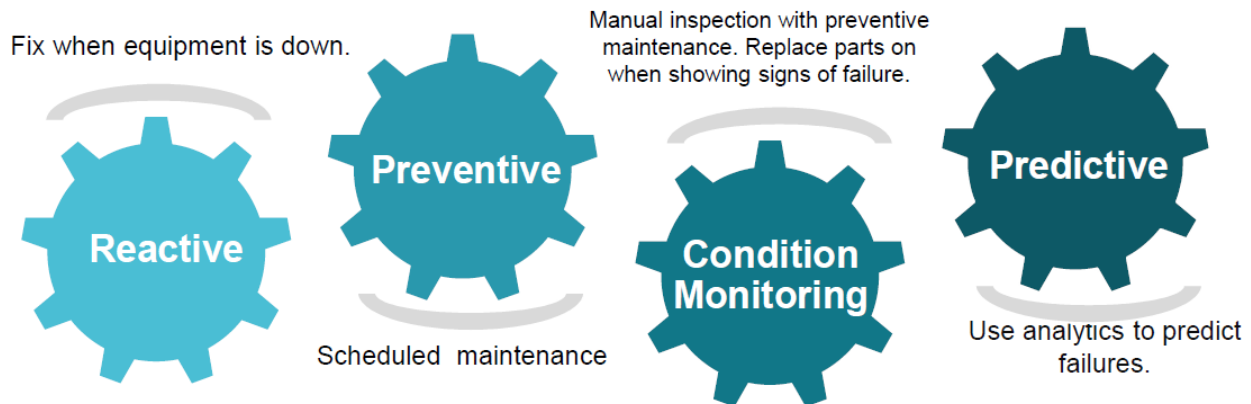


iii. **based Solution**

UCT is one of the early adopters of LoRaWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. **Predictive Maintenance**

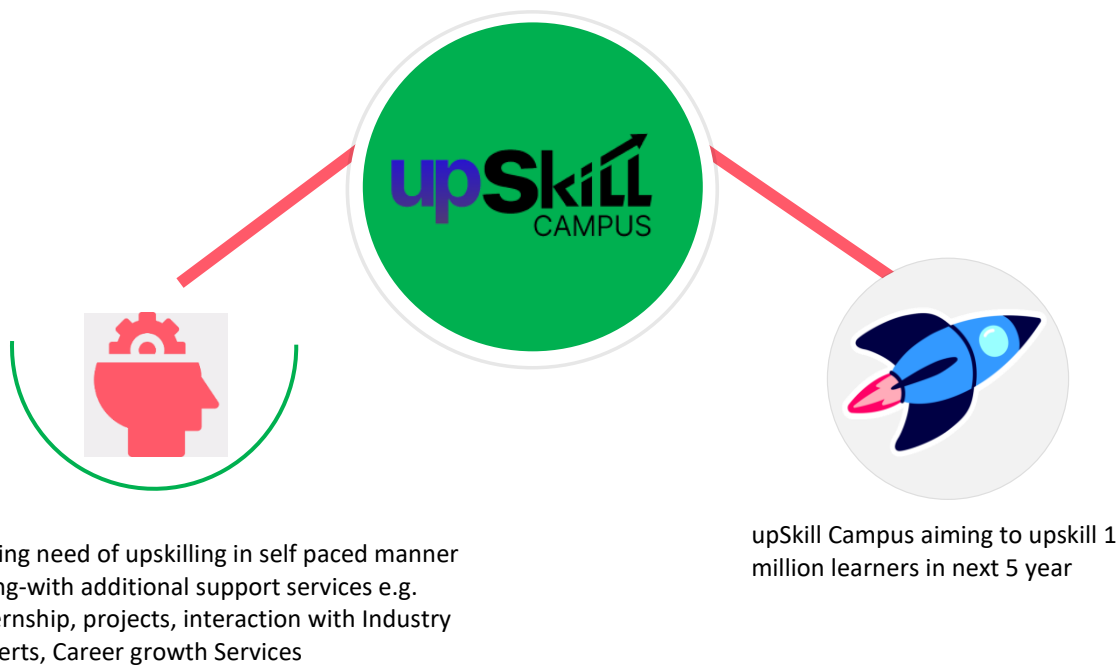
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



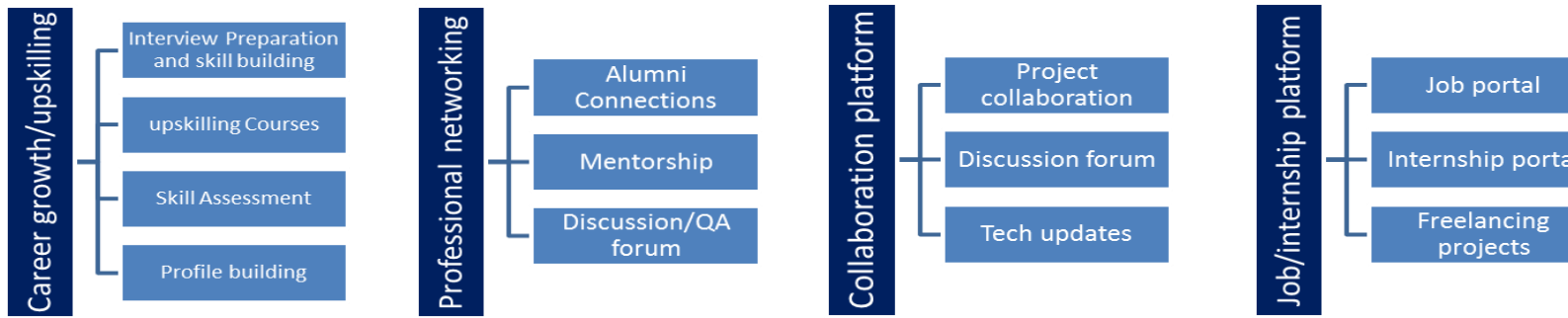
2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The primary objectives of this internship were:

- To gain practical industrial experience in Machine Learning.
- To understand the complete machine learning development lifecycle.
- To work with real agricultural datasets.
- To perform exploratory data analysis and feature engineering.
- To develop predictive models for crop yield and production.
- To compare multiple regression algorithms.
- To evaluate models using standard performance metrics.
- To build an interactive web application using Streamlit.
- To enhance problem-solving and analytical skills.
- To understand the practical challenges involved in deploying AI-based solutions.

2.5 Reference

- Python Documentation
- Scikit-Learn Documentation
- Pandas Documentation
- NumPy Documentation
- Streamlit Documentation
- Agricultural Statistics of India
- UCT Internship Guidelines

2.6 Glossary

Term	Description
ML	Machine Learning
AI	Artificial Intelligence
EDA	Exploratory Data Analysis
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
R ²	Coefficient of Determination
CSV	Comma Separated Values

UI	User Interface
Streamlit	Python framework for web applications
Feature Engineering	Creating new variables from existing data

3. Problem Statement

Agriculture is one of the most important sectors in India, contributing significantly to the country's economy and providing employment to a large portion of the population. However, agricultural productivity is influenced by numerous factors such as crop type, cultivation cost, production cost, cultivated area, season, geographical region, and historical production patterns.

Traditional methods of estimating crop yield and production rely heavily on manual observations and statistical reports, which are often time-consuming, expensive, and prone to inaccuracies. Moreover, farmers and policymakers require timely predictions to make informed decisions regarding crop selection, resource allocation, and agricultural planning.

The primary objective of this project is to develop a machine learning-based prediction system capable of accurately estimating:

- Crop Yield (Quintals per Hectare)
- Future Crop Production (Million Tons)

using historical agricultural data from India.

The developed system aims to assist farmers, researchers, and agricultural organizations by providing data-driven predictions that support improved planning and decision-making.

4. Existing and Proposed solution

4.1 Existing Solutions

Most existing agricultural prediction systems rely on one or more of the following approaches:

- Manual statistical analysis
- Government survey reports
- Basic regression techniques
- Historical trend analysis
- Spreadsheet-based forecasting

Although these methods provide useful information, they have several limitations.

Limitations of Existing Systems

- Low prediction accuracy.
- Inability to handle multiple influencing factors simultaneously.
- Lack of automation.
- No interactive prediction interface.
- Limited scalability for large datasets.
- Minimal feature engineering.
- Poor adaptability to changing agricultural conditions.

These limitations reduce the effectiveness of traditional prediction methods for modern precision agriculture.

4.2 Proposed Solution

To overcome these limitations, a complete machine learning-based prediction system was developed. The proposed solution consists of the following stages:

1. Data Collection

Historical agricultural datasets containing information about crops, cultivation cost, production cost, area, season, region, and production statistics were collected.

2. Data Preprocessing

The collected datasets were cleaned to remove inconsistencies, missing values, and duplicate records.

Categorical variables such as crop names, states, and regions were converted into numerical values using Label Encoding.

Numerical features were standardized using feature scaling to improve model performance.

3. Feature Engineering

Several new features were created to improve prediction accuracy, including:

- Cost Efficiency
- Profitability Index
- Yield-Cost Ratio
- Production Efficiency
- Area-Cost Interaction
- Seasonal Duration
- Production Lag Features
- Rolling Production Average

These engineered features enabled the models to capture complex relationships within the agricultural data.

4. Machine Learning Model Development

Multiple regression algorithms were trained and compared.

The models included:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor
- XGBoost (if available)

Each model was evaluated using standard regression metrics. The best-performing models were selected for deployment.

5. Deployment

A Streamlit-based web application was developed to provide an interactive interface for users.

The application allows users to:

- Select crop data samples
- Predict crop yield
- Predict future production
- Compare actual and predicted values
- Display prediction error

This deployment makes the system accessible to both technical and non-technical users.

4.3 Code submission (Github link)

https://github.com/Pratheeksha26/UpSkillCampus/tree/main/Project4_Ag_Prediction-of-Agriculture-Crop-Production-In-India

4.3.1 Report submission (Github link) :

https://github.com/Pratheeksha26/UpSkillCampus/blob/main/Project4_Ag_Prediction-of-Agriculture-Crop-Production-In-India/Project4Ag_Pratheeksha_USC_UCT.pdf

5. Proposed Design/ Model

The proposed system follows a complete machine learning pipeline consisting of data preprocessing, feature engineering, model training, prediction, and deployment.

The overall architecture ensures efficient processing of agricultural data while maintaining prediction accuracy.

5.1 High Level Diagram

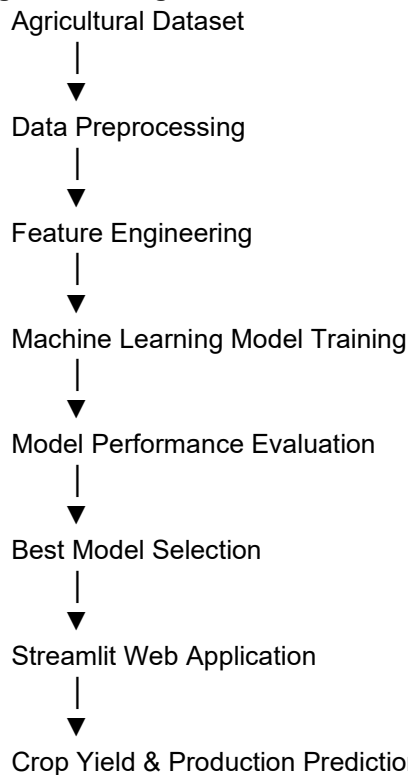


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Interfaces

The project consists of the following major components.

1. Dataset Interface

- Reads CSV datasets.
- Loads historical agricultural information.
- Performs validation checks.

2. Data Processing Module

Responsible for:

- Cleaning data
- Handling missing values
- Encoding categorical variables
- Standardization
- Feature engineering

3. Machine Learning Module

Responsible for:

- Model training
- Hyperparameter tuning
- Performance comparison
- Model selection

4. Prediction Module

Loads trained models and generates predictions based on user-selected inputs.

Outputs include:

- Predicted Crop Yield
- Predicted Crop Production

5. Streamlit User Interface

Provides an easy-to-use graphical interface where users can:

- Select prediction type
- Choose sample data
- Execute predictions
- View prediction accuracy
- Compare actual and predicted values

The application was designed to be intuitive and suitable for demonstration purposes.

6. Performance Test

Performance testing is a crucial phase of any machine learning project as it evaluates the effectiveness, accuracy, and reliability of the developed models. In this project, multiple regression algorithms were trained and evaluated to predict agricultural crop yield and future crop production.

The models were assessed using standard regression evaluation metrics, including:

- **R² Score (Coefficient of Determination):** Measures how well the model explains the variability in the target variable. A value closer to 1 indicates better performance.
- **Mean Absolute Error (MAE):** Represents the average absolute difference between predicted and actual values.
- **Root Mean Square Error (RMSE):** Measures the standard deviation of prediction errors and penalizes larger errors more heavily.

The project involved separate prediction tasks for crop yield and crop production, with different models selected based on their performance.

6.1 Test Plan/ Test Cases

The developed system was tested under various scenarios to ensure its accuracy, reliability, and robustness.

Test Case 1 – Yield Prediction

Test Parameter	Description
Objective	Predict crop yield using historical agricultural data
Input	Crop, State, Region, Cultivation Cost, Production Cost, Area, Season Duration
Expected Output	Predicted Yield (Quintals per Hectare)
Result	Successfully predicted crop yield

Test Case 2 – Production Prediction

Test Parameter	Description
Objective	Predict future crop production
Input	Historical production values, cultivated area, yield, cost, year
Expected Output	Predicted Production (Million Tons)
Result	Successfully predicted future production

Test Case 3 – Streamlit Application Testing

Test Parameter	Description
Objective	Verify functionality of the deployed application
Input	User-selected sample records
Expected Output	Display predicted values and comparison metrics
Result	Successfully displayed predictions and evaluation metrics

Test Case 4 – Model Loading

Test Parameter	Description
Objective	Load saved machine learning models
Expected Output	Models loaded without errors
Result	Successful

Test Case 5 – Feature Processing

Test Parameter	Description
Objective	Validate preprocessing pipeline
Expected Output	Correct feature scaling and encoding
Result	Successful

6.2 Test Procedure

The following steps were followed during model evaluation:

1. Historical agricultural datasets were collected and merged.
2. Data cleaning was performed to remove inconsistencies and handle missing values.
3. Feature engineering techniques were applied to generate additional informative features.
4. Categorical variables were encoded using Label Encoding.
5. Numerical features were standardized using feature scaling.
6. The dataset was divided into training and testing sets.
7. Multiple machine learning regression algorithms were trained.
8. Each model was evaluated using R^2 Score, MAE, and RMSE.
9. The best-performing models were selected and saved.
10. The trained models were integrated into a Streamlit application for interactive prediction.

6.3 Performance Outcome

Crop Yield Prediction Results

Several regression algorithms were evaluated for crop yield prediction. Among all models, the **Decision Tree Regressor** demonstrated the highest prediction accuracy.

Best Model: Decision Tree Regressor

Performance Metrics:

Metric	Value
R^2 Score	0.999
Mean Absolute Error (MAE)	5.15 Quintals/Hectare
Root Mean Square Error (RMSE)	9.64 Quintals/Hectare

The exceptionally high R^2 score indicates that the model accurately captured the relationship between agricultural input features and crop yield.

Crop Production Prediction Results

Multiple regression models were also trained for predicting future crop production. The **Random Forest Regressor** provided the best performance among the evaluated models.

Best Model: Random Forest Regressor

Performance Metrics:

Metric	Value
R^2 Score	0.085
Mean Absolute Error (MAE)	131.07 Million Tons
Root Mean Square Error (RMSE)	389.54 Million Tons

Although production prediction proved more challenging due to the complexity of agricultural production patterns and the limited dataset, the Random Forest model outperformed the other algorithms evaluated.

7. My learnings

This internship provided valuable exposure to the practical application of Machine Learning and Data Analytics in solving real-world agricultural problems. Throughout the project, I gained experience in handling datasets, developing predictive models, and deploying machine learning applications.

Technical Skills Acquired

- Data preprocessing and cleaning using Pandas.
- Exploratory Data Analysis (EDA).
- Feature engineering for improving model performance.
- Regression model development using Scikit-learn.
- Model evaluation using statistical performance metrics.
- Hyperparameter tuning and model comparison.
- Model serialization using Joblib and Pickle.
- Interactive application development using Streamlit.
- Version control and project organization using GitHub.

Professional Skills Developed

- Problem-solving and analytical thinking.
- Project planning and execution.
- Technical documentation.
- Time management.
- Independent learning and research.
- Debugging and troubleshooting.
- Presentation and communication skills.

This internship enhanced my confidence in implementing end-to-end machine learning projects and strengthened my understanding of real-world AI applications.

8. Future work scope

Although the developed system provides accurate predictions for crop yield and reasonable predictions for crop production, several improvements can be made in future work to further enhance its capabilities.

Future Enhancements

- Incorporate real-time weather data from online APIs.
- Integrate soil health and nutrient information.
- Include satellite imagery for precision agriculture.
- Utilize deep learning models such as LSTM and Transformer-based architectures for time-series forecasting.
- Expand the dataset to include more crops, states, and recent agricultural records.
- Implement automated hyperparameter optimization.
- Develop a multilingual user interface for wider accessibility.
- Deploy the application on cloud platforms such as AWS, Azure, or Google Cloud.
- Build a mobile application for farmers.
- Integrate recommendation systems for crop selection and fertilizer usage.
- Connect the application with IoT sensors for real-time field monitoring.

By incorporating these enhancements, the system can evolve into a comprehensive intelligent agricultural decision-support platform, assisting farmers, researchers, and policymakers in improving productivity and promoting sustainable farming practices.